

WHY We Are Doing This Research

Scientific motivation, gap, controls, and value

Cross-Cycle Explanation Drift in Solar Flare Forecasting

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Status	REVIEWED / BASELINE LOCKED
Research domain	Space Weather Machine Learning Explainable AI

1. Core Motivation

WHY

A forecasting model can preserve similar aggregate skill while the pattern of feature attribution observed on a new temporal regime changes. Performance-only evaluation can therefore miss a meaningful change in model behavior.

2. Why Cycle 25 Is a Useful Stress Test

Cycle 25 is a later solar-activity regime with potentially different feature distributions, flare prevalence, active-region populations, and operating conditions. For this study it is treated as a cross-cycle temporal/domain-shift evaluation. We do not automatically label it "out of distribution" until support diagnostics justify that stronger term.

Potential shift	Why it matters
Feature/covariate shift	The model may see magnetic-feature ranges/combinations that differ from Cycle 24.
Label-prevalence shift	A different $\geq M$ flare rate can affect predictive metrics and aggregate SHAP summaries.
AR-population shift	The mix of active-region complexes can differ between periods.
Quality/missingness shift	Data availability or processing differences can create artificial drift.
Solar-cycle phase effects	Prior work shows forecasting behavior can vary with solar-cycle phase, so same-cycle temporal controls are needed.

3. Why Prediction Metrics Alone Are Not Enough

TSS answers whether categorical forecast skill transfers. It does not tell us whether the model assigns importance to features in a similar way across evaluation regimes. SHAP adds a second diagnostic axis: model attribution.

If we only inspect...	What we can miss
TSS / F1 / HSS	Attribution redistribution while aggregate skill remains similar.
Global SHAP rank only	Sampling noise, prevalence differences, correlated-feature rank swaps, and local heterogeneity.
Raw Cycle-24 vs Cycle-25 SHAP	Normal same-cycle temporal variability and covariate-support differences.

4. The Defensible Research Gap

Where This Study Adds Value

The novelty is the controlled combination, not any single component

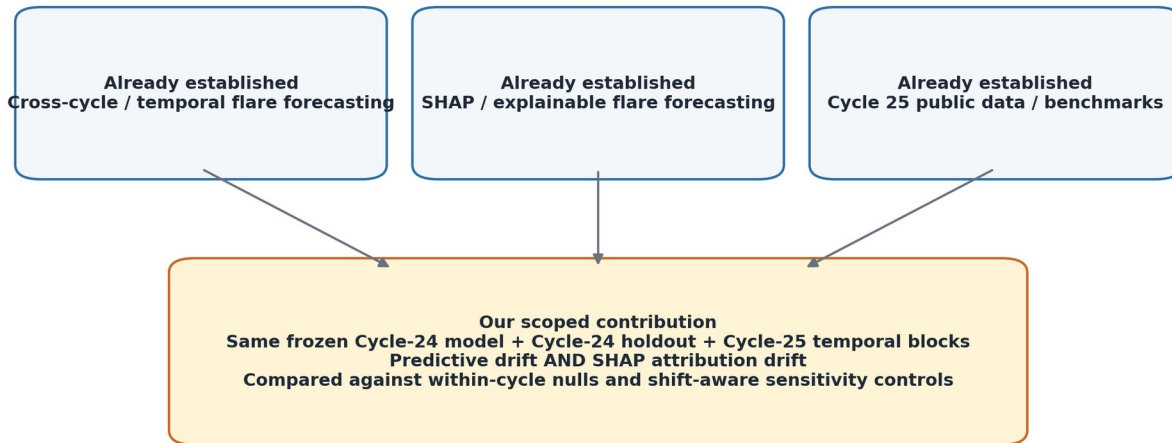


Figure 1. Literature components already exist; the controlled combination is the targeted gap.

Scoped novelty

Based on the literature reviewed for the current design, the targeted contribution is not cross-cycle prediction itself and not SHAP itself. It is the controlled comparison of predictive drift and explanation drift for the same frozen Cycle-24 model, benchmarked against within-cycle nulls and shift-aware robustness checks. This novelty statement must be re-verified immediately before submission.

5. Why the Null Baseline Must Include Two Controls

Control	Purpose
Entity-level resampling / bootstrap inside Cycle 24	Measures ordinary sampling variability at matched sample sizes.
Temporal Cycle-24 blocks	Measures how much attribution can change across time/solar-cycle phase even without crossing into Cycle 25.

Using both controls is stronger than a single bootstrap null because solar forecasting itself can vary with solar-cycle phase.

6. Why Shift-Aware Sensitivity Is Needed

A model is frozen, so a changed SHAP distribution may arise because the Cycle-25 input distribution changed, not because the model parameters changed. The study therefore separates the deployment-style observation from sensitivity analyses that reduce obvious distributional confounding.

Analysis layer	Question answered
Observed-distribution comparison	What attribution pattern is seen on the real Cycle-25 sample?
Class-conditional SHAP	Does drift persist within positive and negative label groups?
Prevalence-standardized analysis	Would the conclusion change if class mix were standardized?
Common-support / reweighting sensitivity	Does the result persist when comparing more comparable input regions?
Label-source sensitivity (if feasible)	Could event-catalog differences be driving the result?

7. Why Correlated Features Matter

SHARP parameters are often strongly correlated. Prior work reports correlations as high as about 0.8-0.99 among several groups, which can make individual-feature importance unstable. Therefore, rank swaps among correlated variables cannot be treated as strong evidence by themselves; group-level attribution stability is required.

8. Scientific Value of Every Outcome

Pre-Registered Interpretation Matrix

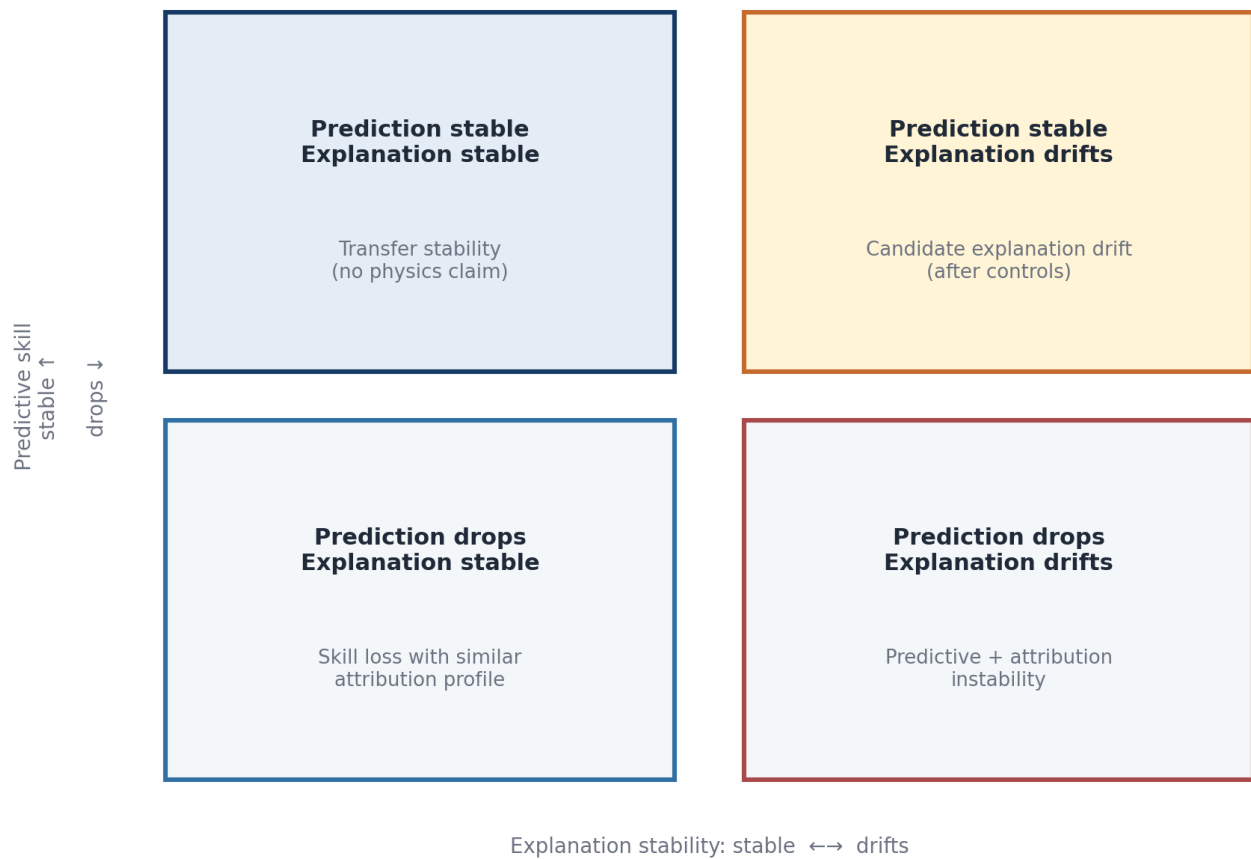


Figure 2. Every pre-defined result pattern produces a useful, bounded scientific conclusion.

9. One-Line WHY

ONE-LINE WHY

We are testing whether a Cycle-24-trained flare model transfers to Cycle 25 not only in predictive skill but also in its observed feature-attribution structure - and whether any attribution change is larger than normal temporal/sampling variability rather than an artifact of data shift.

Reference Basis

These references support the current design. They are not the final literature review; novelty-sensitive claims must be re-checked before submission.

[1] Bobra et al. (2014), SHARP data products and HARP/NOAA mapping, Solar Physics. [Source](#)

[2] Hoeksema et al. (2014), HMI vector magnetic-field pipeline and definitive SHARP products, Solar Physics. [Source](#)

[3] Chen et al. (2019), SHARP flare precursors and correlated-feature instability, Space Weather. [Source](#)

[4] Goodwin et al. (2024), solar-cycle and training-window effects in simulated real-time flare forecasting, ApJ. [Source](#)

[5] Hu et al. (2025), data-source quality issues and Cycle-24-to-25 evaluation, arXiv:2512.13417. [Source](#)

[6] Wu et al. (2026), interpretable major-flare forecasting with SHAP, MNRAS. [Source](#)

[7] Li et al. (2026), explainable operational flare forecasting with SHAP, Space Weather. [Source](#)

[8] Roy et al. (2026), SuryaBench, Scientific Data. [Source](#)

[9] Liu et al. (2026), FlareDB for Solar Cycles 24 and 25, Scientific Data. [Source](#)